Yash Soni

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Education

Wilfrid Laurier University — **Expected Graduation: April 2027**
B.Sc in Data Science | Big Data Concentration | Economics Minor — *Waterloo, ON*

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Statistics, Regression Analysis, Probability, Discrete Mathematics

Experience

Data Scientist Intern — **May 2026 – August 2026**
Manulife Financial — *Toronto, ON*

- Productionized a **Random Forest** model for account takeover fraud detection across **129 features** scoring **7,000+** daily transactions.
- Applied **A/B testing** to validate the Random Forest against a baseline before deployment, increasing flagged fraudulent transactions by **13%**.
- Developed unsupervised **clustering models** to surface internal fraud anomalies across employee behavior in high-dimensional transaction data.
- Built an internal **Streamlit** feature analysis library with configurable outlier detection and **PDF** reporting, reducing model training time by **12%**.
- Built a **LangGraph** Text-to-SQL agent with **Azure AI** semantic indexing, cutting business user table lookup time from 60s to 15s.

Machine Learning Research Assistant — **January 2026 – March 2026**
Wilfrid Laurier University — *Waterloo, ON*

- Developed and evaluated **ML models** for autonomous **UAV decision-making** using 50,000 validated flight scenarios across 10 domains.
- Benchmarked **3 ML architectures** on UAV planning and perception tasks, achieving **81.3% classification accuracy** across 10 domains.
- Evaluated **multi-domain AI reasoning** across navigation, energy management, and ethical decision-making in autonomous **UAV systems**.

Data Engineer Intern — **September 2025 – December 2025**
John Hancock — *Toronto, ON*

- Deployed scalable **Databricks** pipelines using PySpark and SQL for enterprise data ingestion, transformation, and ML analytics workflows.
- Automated **ML-oriented** ingestion and transformation workflows, cutting processing time by **30%** across reporting and modeling systems.
- Designed **PySpark monitoring pipelines** to track data quality, execution reliability, and reporting consistency across analytics systems.
- Modernized legacy **SQL** transformations into distributed PySpark workflows, improving scalability and downstream data accessibility for AI.
- Operationalized validated datasets with analytics and engineering teams to accelerate reporting, modeling, and AI-driven decision support.

Data Analyst Intern — **May 2024 – August 2024**
RBC — *Toronto, ON*

- Validated daily mortgage disbursement operations exceeding **$2.5M** while maintaining accuracy and compliance across regulated workflows.
- Built **Power BI dashboards and Excel trackers** reducing escalation timelines by **20%** for corrupted operational records across teams.
- Collaborated with compliance teams to investigate discrepancies and strengthen data transparency across regulated processes.

Teaching Assistant — **September 2024 – April 2025**
Wilfrid Laurier University — *Waterloo, ON*

- Guided **200+ students** through analytics and calculus coursework by debugging **Python and R** workflows and clarifying core concepts.

Projects

Stratton Oak Bot | *Python, Flask, LangChain, OpenAI API*

- Architected a **retrieval-augmented generation** assistant for financial documents using LangChain retrieval and grounded LLM generation.
- Implemented a **Flask REST API** enabling conversational querying over portfolio metrics, financial statements, and market data indicators.

Cortex, HackCanada 2026 Top 8 | *Next.js, React, TypeScript, Three.js, FastAPI, Python, Google Gemini* — **Live Demo**

- Built Cortex, an AI platform ingesting **GitHub repositories, PDFs, and text** to evaluate skills across **51 computer science categories**.
- Engineered a **FastAPI** pipeline to extract features, score proficiency, and evaluate developer artifacts across multiple input sources.
- Created an interactive **Three.js 3D brain** visualizing skill strength across 9 regions, updating dynamically as users upload new evidence.

Trade-Risk, CxC Data Science Hackathon Finalist | *Python, Flask, TensorFlow, Keras, Pandas, NumPy, Backboard.io, Gemini API, pytest*

- Built a scenario-based tariff risk engine in **Python** to model Canadian export sector exposure under evolving trade policy conditions.
- Processed trade datasets using **Pandas** and **NumPy** to engineer sector-level features and score risk via deterministic economic signals.
- Integrated **TensorFlow/Keras** deep learning models for comparative ML analysis and benchmarking against the rule-based risk scoring engine.
- Deployed a **Flask API** with **Backboard.io** caching to serve consistent, auditable Gemini scenario explanations for production outputs.

Technical Skills

Languages: Python, SQL, R, Java
Platforms & Tools: Azure Databricks, Snowflake, Jupyter, Git, Linux, Power BI, Excel, Streamlit
Libraries & Frameworks: Pandas, NumPy, scikit-learn, PyTorch, TensorFlow, LangChain/LangGraph, Matplotlib, pytest
Certifications: Unsupervised Learning, Recommenders, Reinforcement Learning (Stanford Online), Prompt Engineering (IBM)